GLOBAL AI CONFERENCE · KEYNOTE SESSION

Advanced Generative AI Image Generation Capabilities

From random noise to vivid imagination — how machines learned to paint what we describe, and where the frontier is heading.

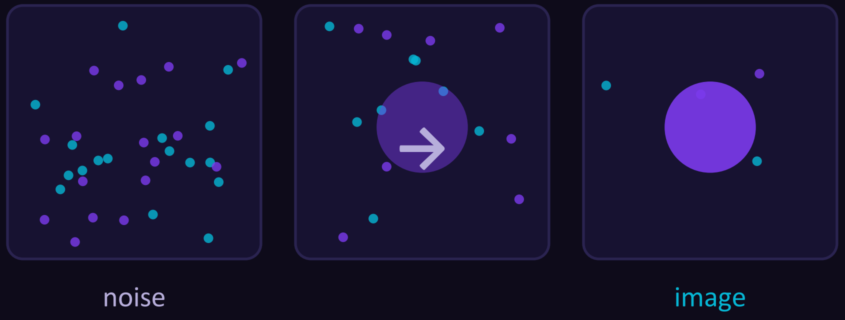
Designed for every level — first-time learners to seasoned practitioners.

AI Art Generator
AI Image Generator
TEXT TO IMAGE
<prompt> a frog in the foreground, in the rain forest, side view, high contrast, vibrant colors, 4k resolution, photo-realistic
IMAGE TO IMAGE
Upload your image, select a style, and transform.
IMAGE DIMENSION
Width x Height of the image
512
512
Generate

WHY THIS MATTERS

You typed a sentence. The machine painted it.

A decade ago, turning a phrase into a photorealistic image needed an artist and days of work. Today it takes a sentence and a few seconds. That shift is not a better paintbrush — it is a fundamentally new way for machines to understand and create.



~30s from prompt to a 4K image on today's frontier models

3 questions any image model must answer — the spine of this talk

Text→Pixels now one shared problem, not two separate worlds

THE ONE IDEA TO HOLD ONTO

Three questions every image model must answer

Strip away the jargon and every text-to-image system is solving the same three problems. We will spend the session on each — in this order.

PART 1

How do I imagine?

Turn a blank canvas of pure noise into a coherent picture. This is the diffusion engine.

PART 2

How do I understand words?

Map a sentence and a picture into the same space so 'meaning' can steer the brush. This is embeddings & CLIP.

PART 3

What can I actually do?

Create, edit, upscale, restyle — and where the frontier is heading in 2026.

ROADMAP

Where we are going

The Engine — Diffusion

01

Noise, denoising, latent space, and the Stable Diffusion pipeline

The Translator — Words ↔️ Pixels

02

Embedding spaces, CLIP, contrastive learning, autoencoders

Capabilities & the Frontier

03

What models do today, how architectures evolved, the 2026 landscape

Hands-On & Takeaways

04

Two live demos and the mental model to carry home

PART 1

The Engine: Diffusion

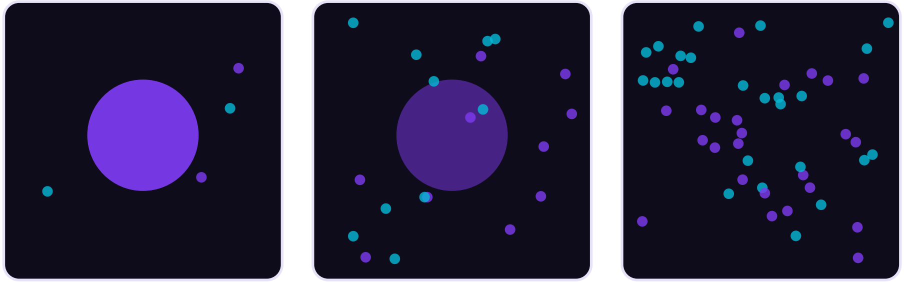
How a model turns pure static into a picture — the single most important idea in modern image generation.

THE CORE TRICK

Sculpting an image out of chaos

Imagine a sculptor staring at a rough block of marble. They don't add stone — they remove what isn't the statue. Diffusion models work the same way. They start from a canvas of pure random noise and, step by step, remove what isn't the picture until an image emerges.

The mental model: Generation is denoising, run in reverse. Teach a network to clean up noisy images, then ask it to clean up nothing but noise — and it hallucinates a brand-new image.

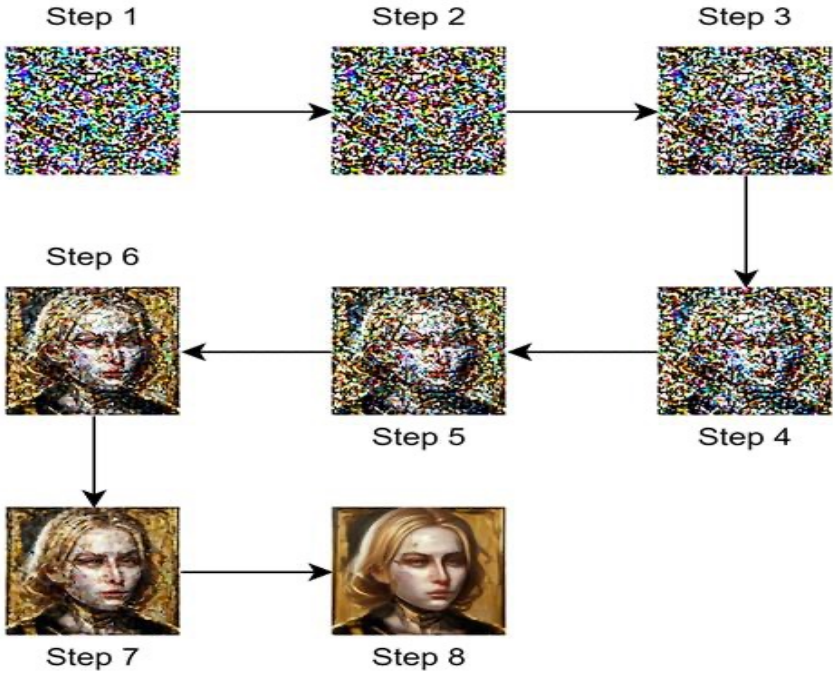


clear image

pure noise

→ add noise (forward / training)

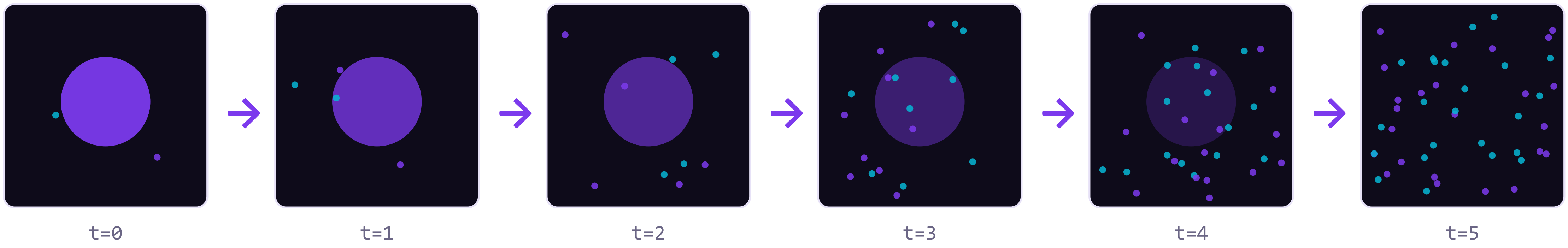
← remove noise (reverse / generate)



STEP 1 OF THE RECIPE

Learning by destroying: the forward process

Before a model can rebuild images, we deliberately wreck them. We take a clean training image and add a tiny bit of random noise — then a little more, then more — over many steps, until nothing is left but static. Crucially, we record exactly how much noise we added at each step.

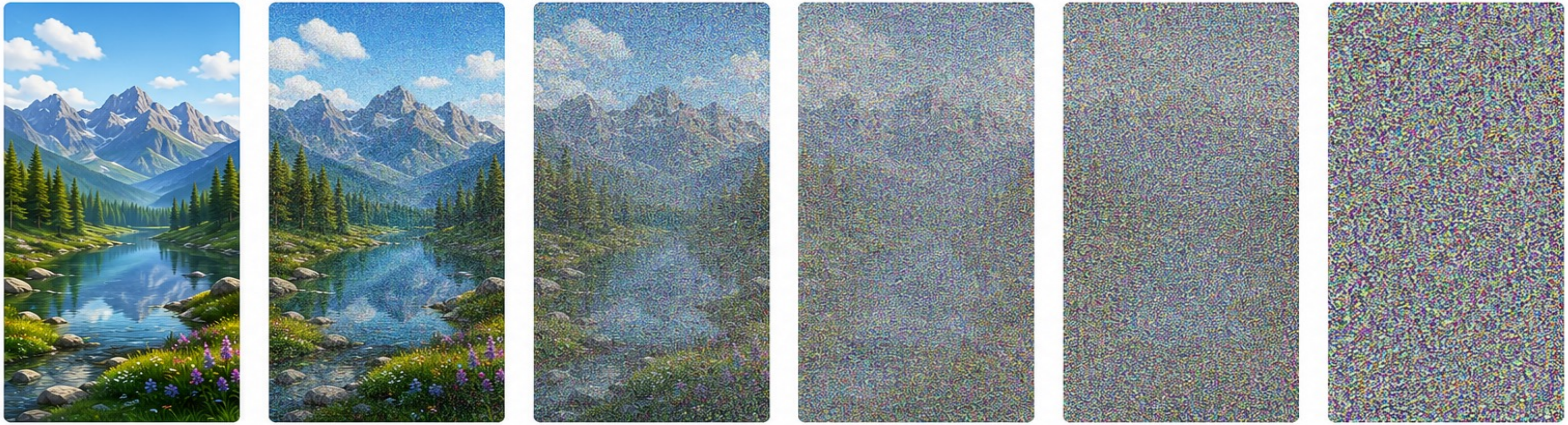


Why bother? Each noisy-vs-cleaner pair becomes a free training example. The model sees millions of them and learns one skill: given a noisy image, predict the noise that was added.

FORWARD NOISING PROCESS

Gradually adding noise to data until it becomes pure random noise

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6
Noise Level: 0%	Noise Level: 20%	Noise Level: 40%	Noise Level: 60%	Noise Level: 80%	Noise Level: 100%



Original Image	Slight noise added, details mostly clear	Moderate noise, some details lost	High noise, details hard to see	Very high noise, image barely recognizable	Pure random noise, no information remains

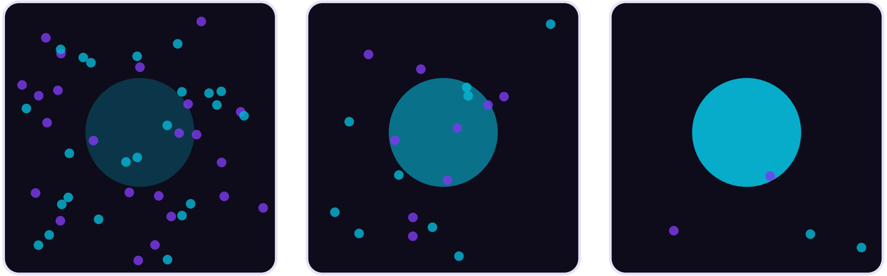
0% (Clean Data) — Increasing Noise → 100% (Pure Noise)

STEP 2 OF THE RECIPE

The real magic: the reverse process

Now flip it. Hand the trained network a frame of pure noise and a prompt. It predicts the noise, subtracts a slice of it, and repeats — dozens of times. Each pass reveals a little more structure, like a photo developing in a darkroom.

1	**1 · Predict**	The network estimates: how much of this is noise, and what shape lies underneath?
2	**2 · Subtract**	Remove a controlled fraction of that predicted noise — never all at once.
3	**3 · Repeat**	Loop the prediction-and-subtract step until a clean image remains.



← fewer noise, more image

The prompt steers every step — that is where words become pictures (Part 2).

REVERSE DIFFUSION: BRINGING IMAGES BACK TO LIFE

Like a photo emerging in the darkroom, reverse diffusion gradually transforms random noise into a coherent image.

START: RANDOM NOISE
Pure randomness, no structure

↓

ITERATIVE REFINEMENT
Each step removes noise and reveals more detail

↓

STRUCTURE EMERGES
Shapes, edges, and patterns become recognizable

↓

IMAGE RESTORED
A clear, coherent image is revealed

AI CONNECTION
Reverse diffusion models learn to undo noise, reconstructing data one step at a time.

THE PROCESS: NOISE TO IMAGE

STEP 1
Random Noise
No structure, pure randomness

STEP 2
Early Structure
Basic patterns begin to form

STEP 3
Shapes Appear
Edges and forms become visible

STEP 4
Details Emerge
Textures and details are refined

STEP 5
Almost There
Image is mostly restored

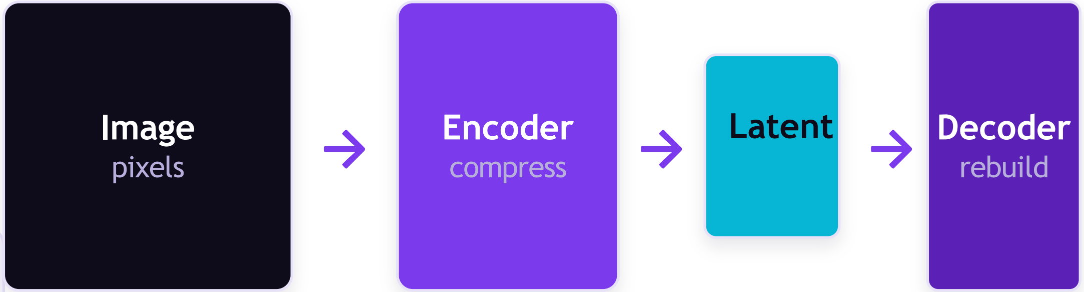
STEP 6
Image Restored
Clear, coherent image achieved

Think in concepts, not pixels: latent space

Denoising a full-resolution image millions of times would be brutally expensive. The breakthrough behind Stable Diffusion was to do the work in a compressed 'latent' space — a small, meaning-rich summary of the image instead of every pixel.

Analogy: a recipe, not the dish

You don't memorize a meal bite-by-bite — you remember the recipe. A latent code is the recipe of an image: far smaller, but enough to reconstruct the dish. Diffusion runs on the recipe, then a decoder cooks the full picture.



Diffusion happens HERE — in the small latent

~48× smaller to process

COMPRESSION & RECONSTRUCTION

Complex data can be compressed into compact representations
and reconstructed with high fidelity

ORIGINAL DATA (HIGH INFORMATION)

COMPRESS

Reduce data to essential information

COMPRESSED REPRESENTATION (LOWER INFORMATION)

HERB ROASTED SALMON

with Asparagus & Lemon Cream

30 min | 2 servings | Medium

INGREDIENTS

- 2 salmon fillets
- 1 bunch asparagus
- 1 cup cherry tomatoes
- 1/2 cup heavy cream
- 1 lemon
- 2 tbsp olive oil
- 2 garlic cloves
- Salt & pepper
- Fresh herbs (thyme, parsley)

STEPS

1. Preheat oven to 400°F (200°C).
2. Season salmon with salt, pepper, and herbs.
3. Toss asparagus and tomatoes with olive oil, salt, and pepper.
4. Roast salmon, asparagus, and tomatoes for 12–15 min.
5. Make lemon cream: sauté garlic, add cream and lemon zest.
6. Serve salmon over cream sauce with roasted vegetables.

RECONSTRUCTED OUTPUT (HIGH FIDELITY)

RECONSTRUCT

Rebuild high-fidelity output from compressed information

KEY TAKEAWAY

Compression captures the essence.
Reconstruction restores the experience.

PUTTING IT TOGETHER

The Stable Diffusion engine, assembled

Three components, one pipeline. A text encoder turns your prompt into meaning; a denoiser (the U-Net or, today, a Transformer) sculpts the latent guided by that meaning; a decoder turns the finished latent into pixels.

Text encoder

"a red fox in snow" → meaning vector

→

Denoiser (U-Net / DiT)

iteratively cleans the noisy latent, steered by the prompt

↻ repeated ~20–50 steps

→

Decoder (VAE)

latent → full-resolution image

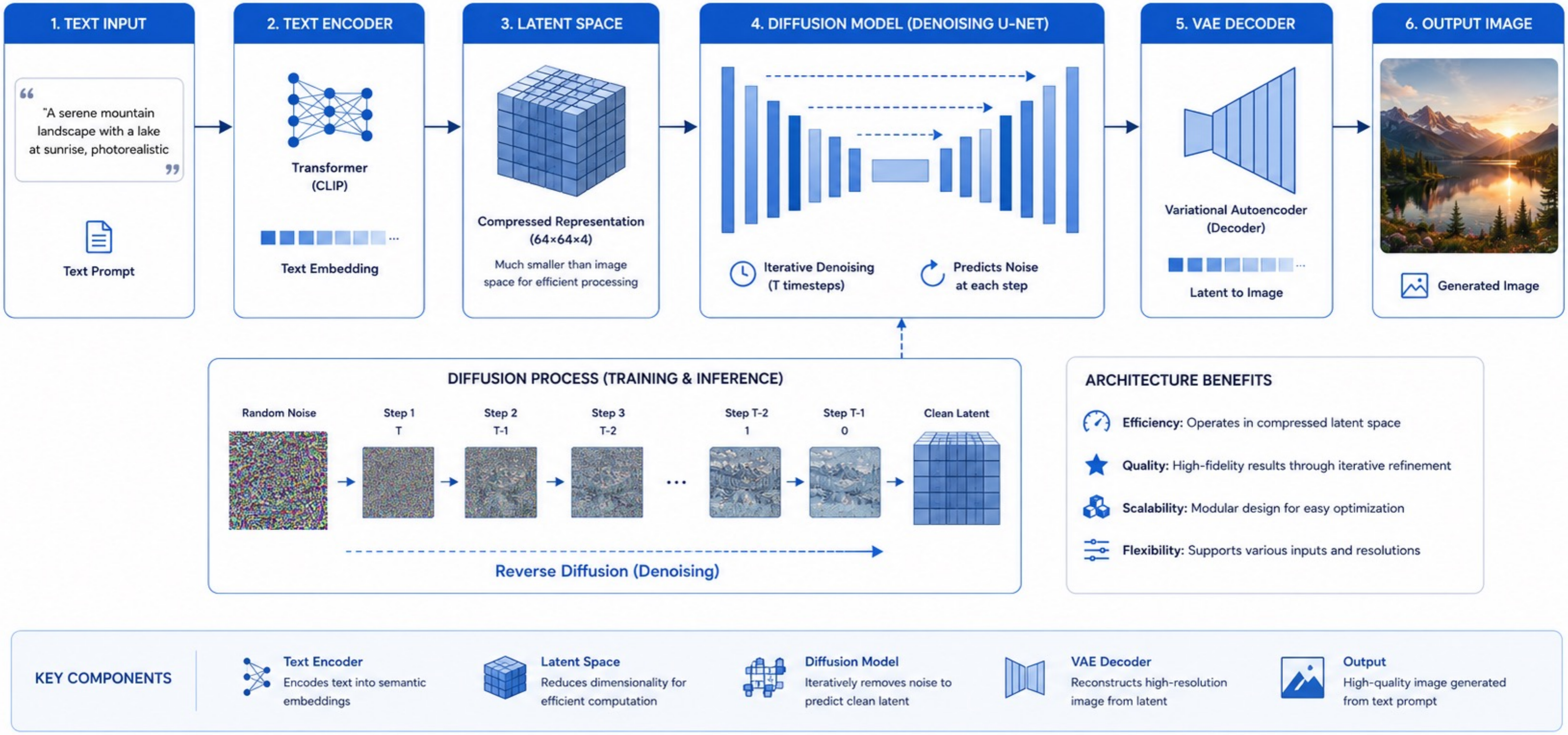
→

Image

Open-source matters: because weights and code are public, this exact pipeline spawned an entire ecosystem — fine-tunes, LoRAs, ControlNets, and the FLUX / SDXL families we'll meet in Part 3.

MODERN DIFFUSION MODEL ARCHITECTURE

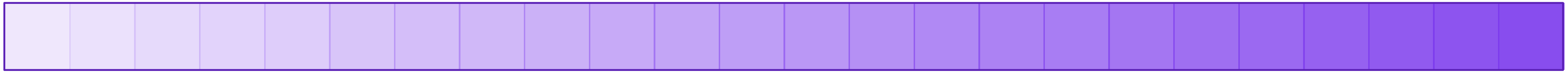
Transforming Text Prompts into High-Quality Images



THE ONE DIAL THAT CHANGES EVERYTHING

Denoising strength: your creative throttle

When you edit or transform an existing image, one parameter decides how far the model strays from the original: denoising strength. Too low and nothing changes; too high and the original is lost. The art is finding the balance.



0.0 1.0

Low (0.2-0.4)

Subtle touch-ups. Keeps composition, colour, identity. Great for restoration.

Mid (0.5-0.7)

Reinterpret while honouring the source. The sweet spot for style transfer.

High (0.8-1.0)

Bold reinvention. The prompt dominates; the original is just a seed.

MAKING IT FAST

From 1000 steps to a handful

Early diffusion models (DDPM) needed up to a thousand tiny denoising steps — accurate but painfully slow. DDIM and later samplers showed you can take bigger, smarter strides and reach near-identical quality in a fraction of the steps.

DDPM

~1000 small steps · slow

DDIM +

~20–50 big steps · fast & deterministic

Why you should care: fewer steps = lower cost and real-time interaction. This single advance is what turned diffusion from a research curiosity into a product you can ship.

PART 2

The Translator: Words ↔️ Pixels

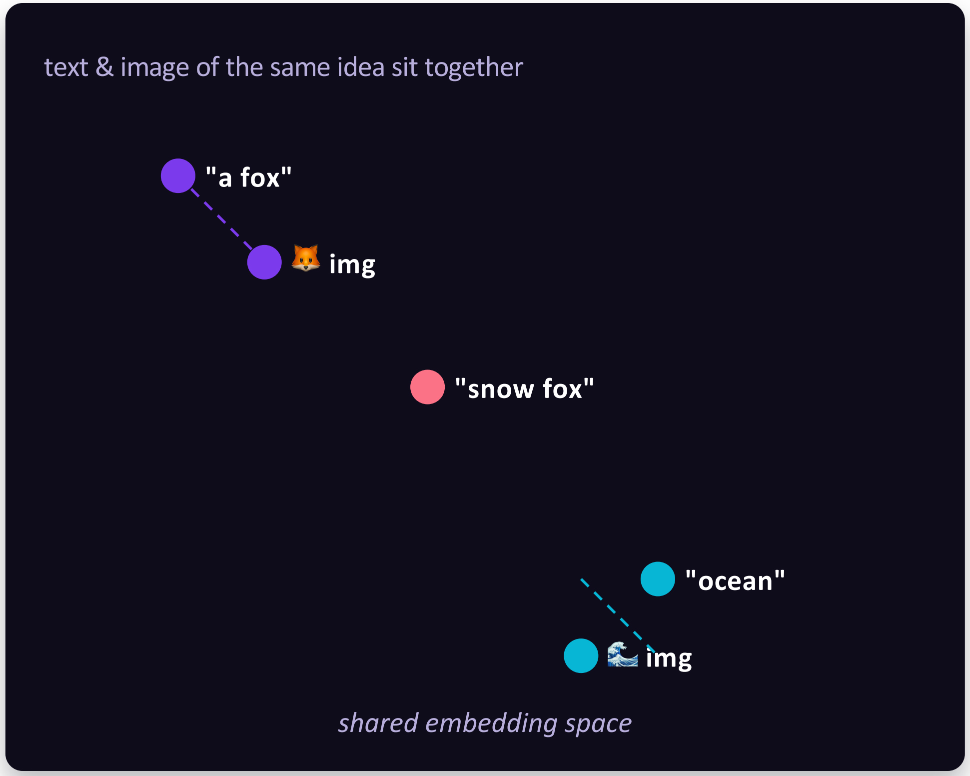
The engine can sculpt — but how does a sentence steer the chisel? The answer is shared meaning.

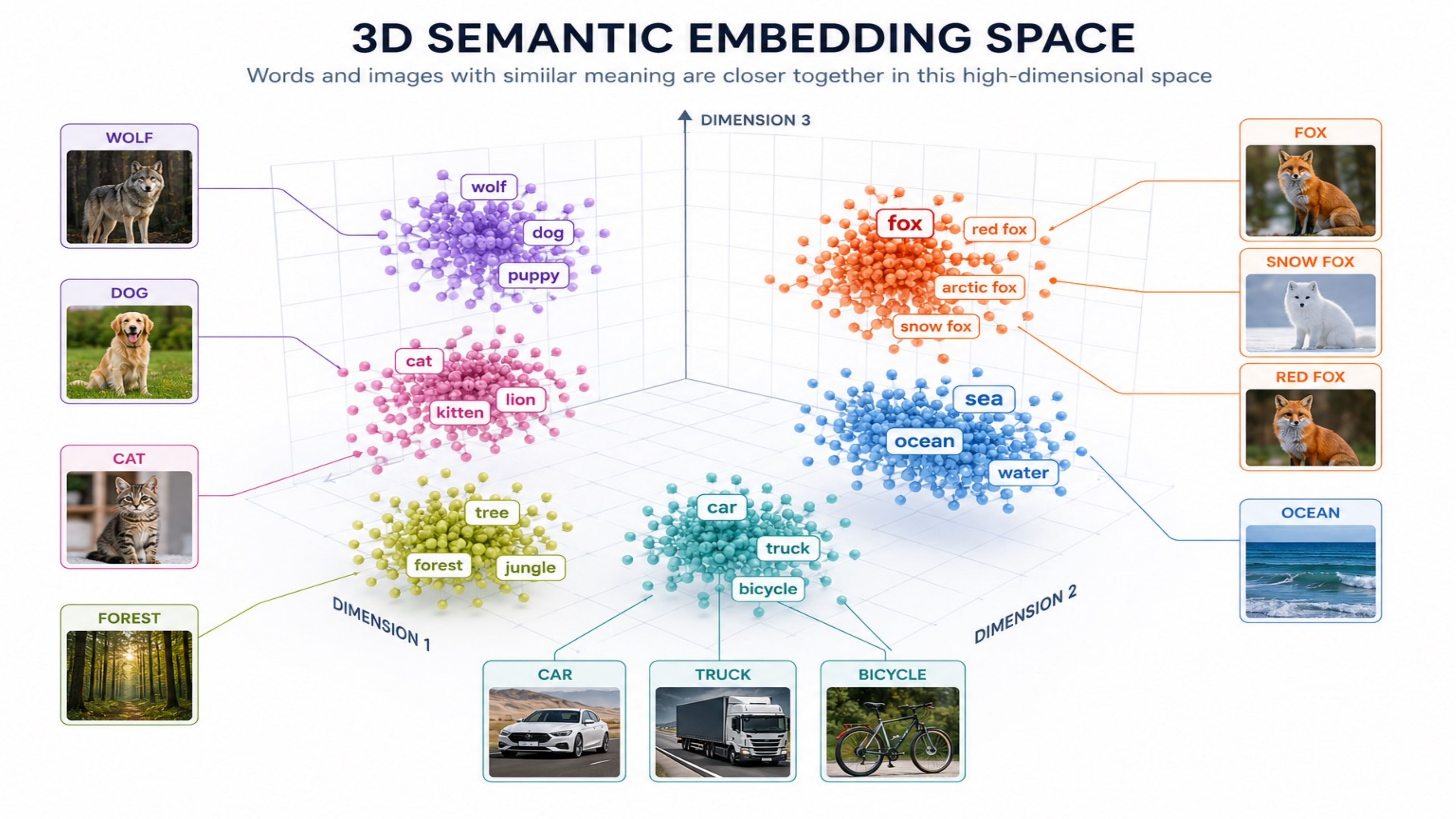
THE BIG IDEA OF PART 2

A shared map for words and pictures

Computers don't understand 'fox' or recognise a photo directly. They turn both into vectors — long lists of numbers. The trick: train so that a picture of a fox and the word 'fox' land in nearly the same spot on one shared map. Closeness on the map means closeness in meaning.

Why it's powerful: once text and images share a map, you can search images with words, caption pictures, and — critically — let a prompt pull the diffusion process toward the right region of image-space.





3D SEMANTIC EMBEDDING SPACE
Words and images with simiilar meaning are closer together in this high-dimensional space
DIMENSION 3
DIMENSION 1
DIMENSION 2
WOLF
DOG
CAT
FOREST
FOX
SNOW FOX
RED FOX
OCEAN
CAR
TRUCK
BICYCLE
wolf
dog
puppy
cat
lion
kitten
tree
forest
jungle
car
truck
bicycle
fox
red fox
arctic fox
snow fox
sea
ocean
water

CLIP: teaching text and images one language

CLIP learns the shared map by studying hundreds of millions of image–caption pairs from the web. It runs each image through an image encoder and each caption through a text encoder, then nudges matching pairs together and mismatched pairs apart.

- **Two encoders, one goal**

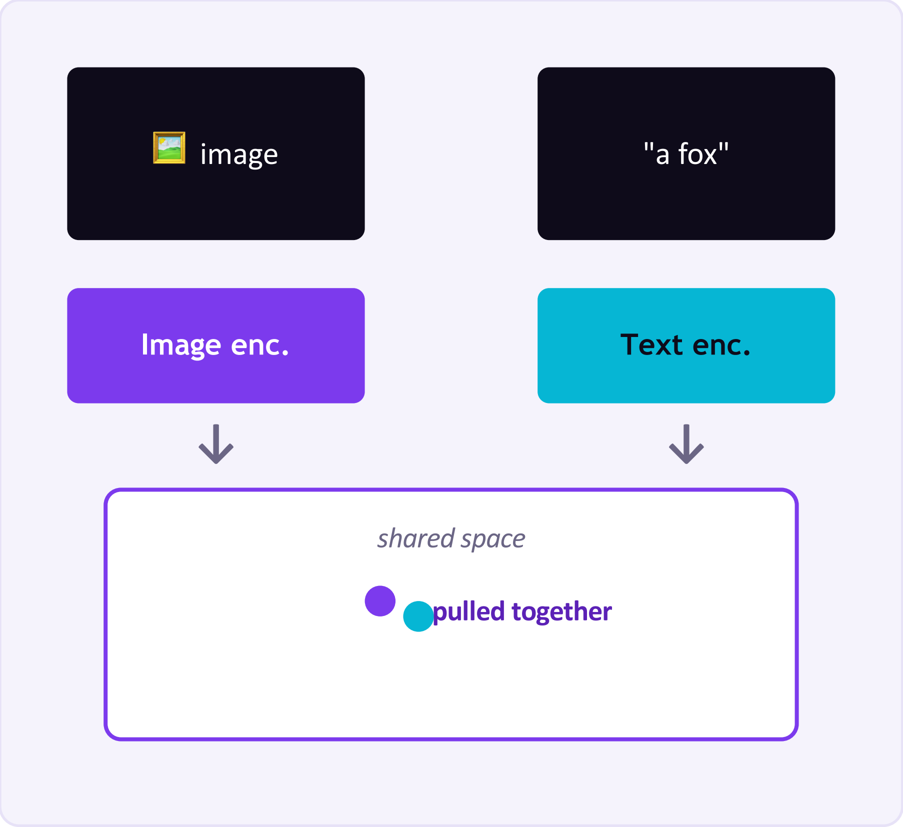
  Image encoder and text encoder produce vectors in the same space.

- **Match vs. mismatch**

  Correct image–caption pairs are pulled close; wrong pairs are pushed away.

- **No hand-labels needed**

  The web's alt-text is the supervision — so it scales to internet size.



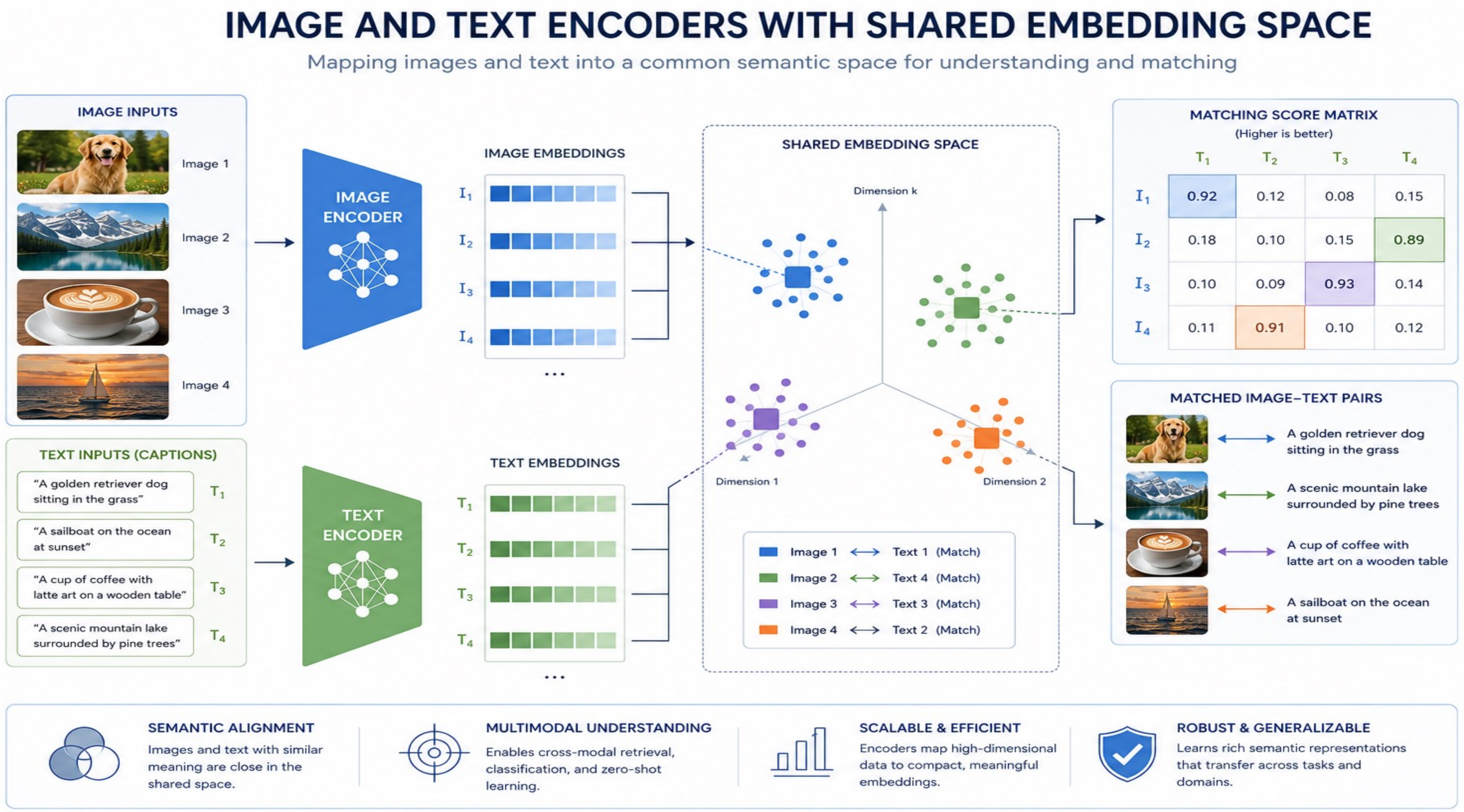


IMAGE AND TEXT ENCODERS WITH SHARED EMBEDDING SPACE
Mapping images and text into a common semantic space for understanding and matching
IMAGE INPUTS
Image 1
Image 2
Image 3
Image 4
IMAGE ENCODER
IMAGE EMBEDDINGS
I1
I2
I3
I4
...
TEXT INPUTS (CAPTIONS)
"A golden retriever dog sitting in the grass" T1
"A sailboat on the ocean at sunset" T2
"A cup of coffee with latte art on a wooden table" T3
"A scenic mountain lake surrounded by pine trees" T4
TEXT ENCODER
TEXT EMBEDDINGS
T1
T2
T3
T4
...
SHARED EMBEDDING SPACE
Dimension k
Dimension 1
Dimension 2
Image 1 ⟷ Text 1 (Match)
Image 2 ⟷ Text 4 (Match)
Image 3 ⟷ Text 3 (Match)
Image 4 ⟷ Text 2 (Match)
MATCHING SCORE MATRIX
(Higher is better)
T1 T2 T3 T4
I1 0.92 0.12 0.08 0.15
I2 0.18 0.10 0.15 0.89
I3 0.10 0.09 0.93 0.14
I4 0.11 0.91 0.10 0.12
MATCHED IMAGE–TEXT PAIRS
A golden retriever dog sitting in the grass
A scenic mountain lake surrounded by pine trees
A cup of coffee with latte art on a wooden table
A sailboat on the ocean at sunset
SEMANTIC ALIGNMENT
Images and text with similar meaning are close in the shared space.
MULTIMODAL UNDERSTANDING
Enables cross-modal retrieval, classification, and zero-shot learning.
SCALABLE & EFFICIENT
Encoders map high-dimensional data to compact, meaningful embeddings.
ROBUST & GENERALIZABLE
Learns rich semantic representations that transfer across tasks and domains.

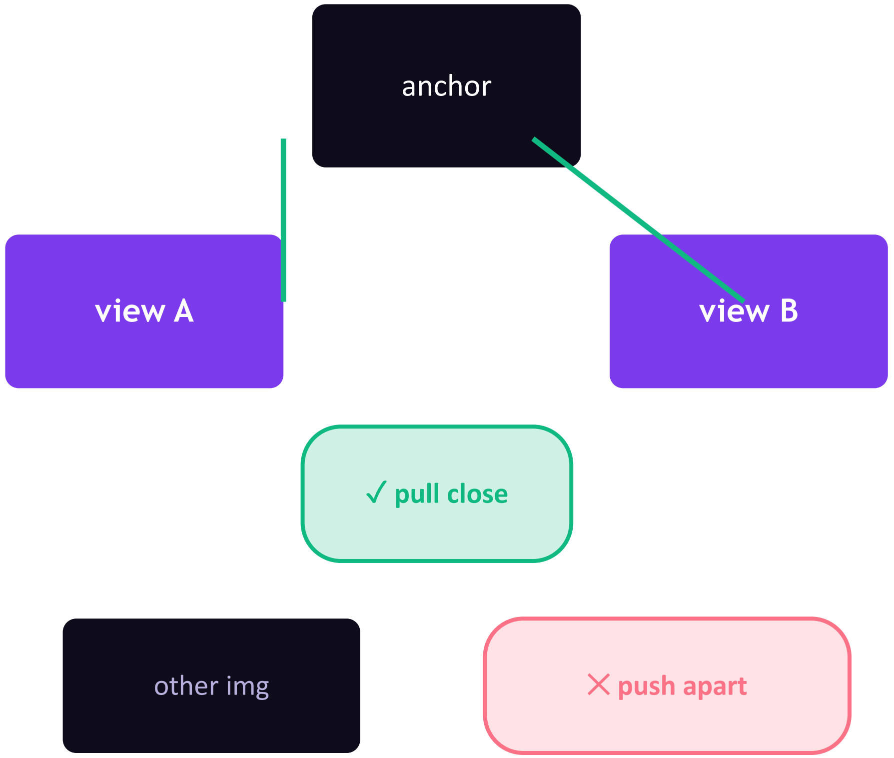
Contrastive learning: understanding by comparison

How do you teach meaning without labels? Show the model what is alike and what is different. Take one image, create two altered views of it (crop, flip, recolour) — those are a positive pair and should match. Every other image is a negative — it should not.

Pull together the two views of the same image.

Push apart views of different images. Repeat billions of times.

SimCLR made this famous; the same family powers CLIP. The payoff: rich features learned from unlabeled data.



THE COMPRESSION BACKBONE

Autoencoders & VAEs: meaningful compression

Remember the encoder/decoder from latent space? That's an autoencoder — a network that squeezes data down to its essence and rebuilds it. Variational autoencoders (VAEs) go further: they organise the compressed space so it's smooth and samplable, which is exactly what generation needs.

Autoencoder

Encoder compresses → bottleneck → decoder reconstructs. Learns what to keep and what to drop.

VAE

Adds structure so nearby points decode to similar images — a space you can smoothly sample and interpolate.

Why it matters here

The VAE is the encoder/decoder of Stable Diffusion — and contrastive learning sharpens what those embeddings mean.

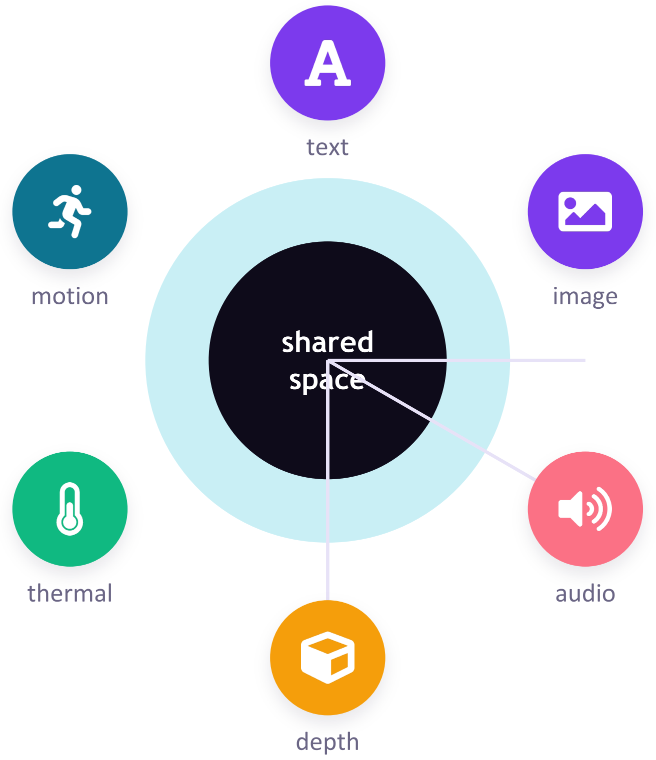
WHERE THIS IS HEADING

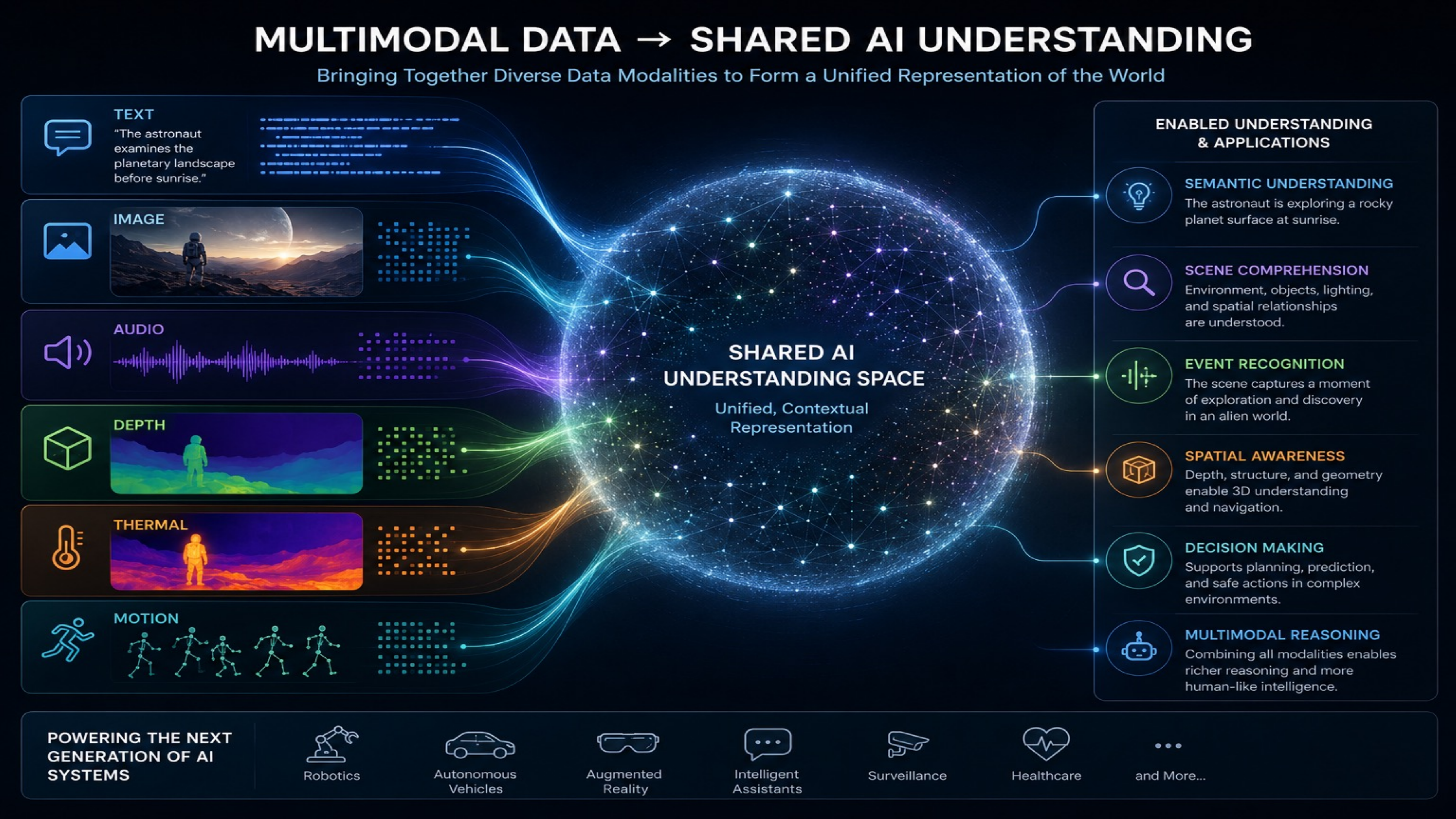
Beyond text + image: one space for everything

If text and images can share a map, why stop there? Research like ImageBind binds many modalities — audio, depth, thermal, motion — into a single embedding space, often without paired data for every combination. The result: search and generate across senses.

What this unlocks

- Hum a tune → get a matching image
- Search a photo library with a sound
- Compose meaning with arithmetic: 'dog' + 'bark' ≈ barking dog
- Cross-modal retrieval & generation in one model




MULTIMODAL DATA → SHARED AI UNDERSTANDING
Bringing Together Diverse Data Modalities to Form a Unified Representation of the World
TEXT
"The astronaut examines the planetary landscape before sunrise."
IMAGE
AUDIO
DEPTH
THERMAL
MOTION
SHARED AI UNDERSTANDING SPACE
Unified, Contextual Representation
ENABLED UNDERSTANDING & APPLICATIONS
SEMANTIC UNDERSTANDING
The astronaut is exploring a rocky planet surface at sunrise.
SCENE COMPREHENSION
Environment, objects, lighting, and spatial relationships are understood.
EVENT RECOGNITION
The scene captures a moment of exploration and discovery in an alien world.
SPATIAL AWARENESS
Depth, structure, and geometry enable 3D understanding and navigation.
DECISION MAKING
Supports planning, prediction, and safe actions in complex environments.
MULTIMODAL REASONING
Combining all modalities enables richer reasoning and more human-like intelligence.
POWERING THE NEXT GENERATION OF AI SYSTEMS
Robotics
Autonomous Vehicles
Augmented Reality
Intelligent Assistants
Surveillance
Healthcare
and More...

PART 3

Capabilities & the Frontier

From neat demos to production tools — what these systems do today, and the 2026 architecture shift you should know.

WHAT THEY DO TODAY

Six capabilities you can use now

Text → Image

Generate original visuals from a prompt.

Inpainting

Replace or fix part of an image seamlessly.

Outpainting

Extend a picture beyond its borders.

Super-resolution

Upscale and sharpen low-res images.

Image → Image

Restyle or transform an existing photo.

Controlled gen

Guide structure with poses, edges, depth.

HOW WE GOT HERE

The backbone evolved: U-Net → DiT → MMDiT

The denoiser at the heart of diffusion has changed shape. Early models used a U-Net; the field then borrowed the Transformer — the same architecture behind LLMs — and finally fused image and text tokens into one attention stream.

2022

U-Net

Convolutional denoiser. Powered SD 1.x / 2.x. Reliable but harder to scale.

e.g. SD 1.5

→

2023–24

Diffusion Transformer (DiT)

Swap convolutions for Transformer blocks. Scales cleanly with data and compute.

e.g. SD3, PixArt

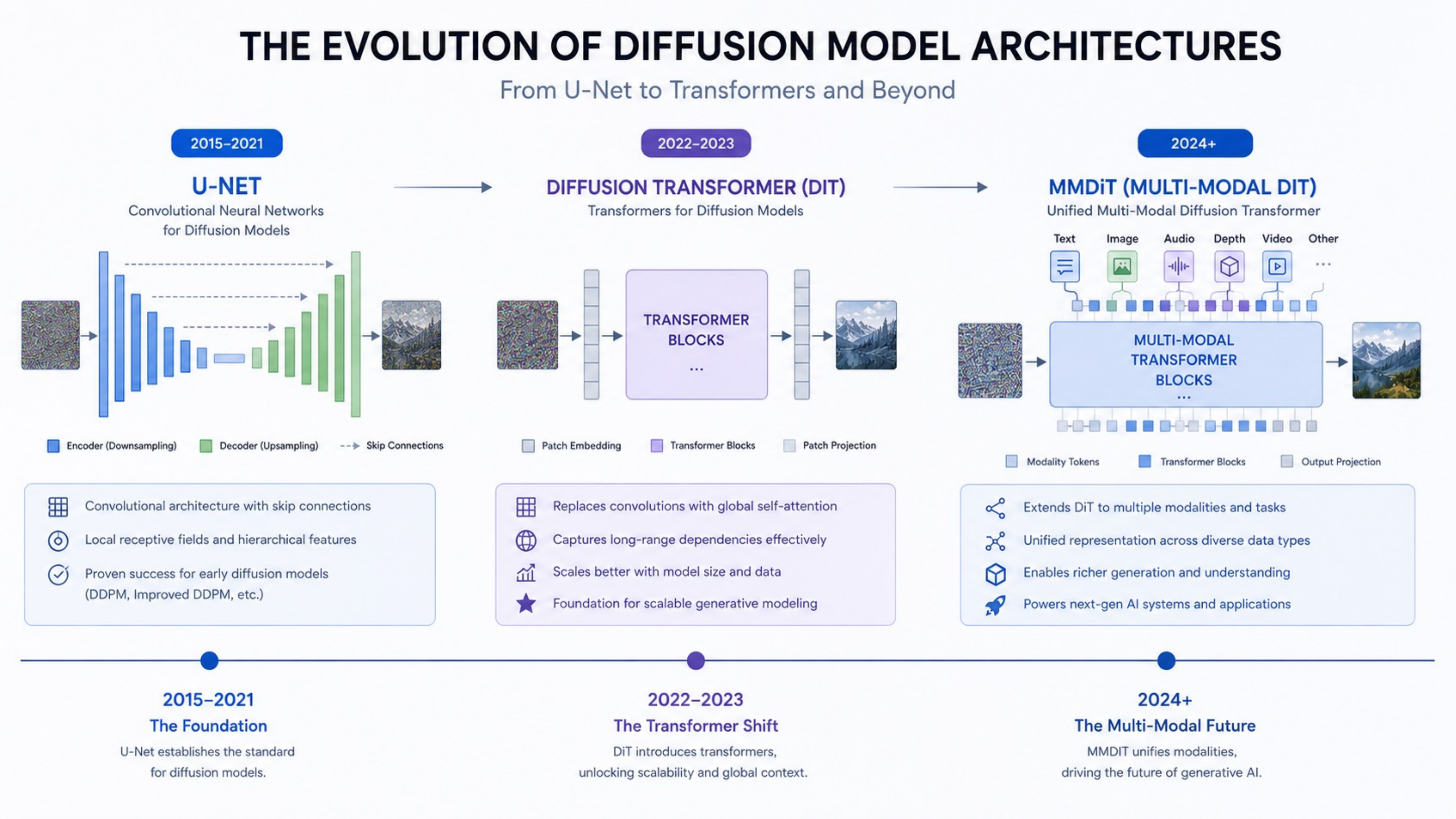
→

2024–26

MMDiT + Flow

Joint attention over image AND text tokens, trained with rectified flow for sharper alignment.

e.g. FLUX, Qwen-Image


THE EVOLUTION OF DIFFUSION MODEL ARCHITECTURES
From U-Net to Transformers and Beyond
2015–2021
U-NET
Convolutional Neural Networks for Diffusion Models
Encoder (Downsampling)
Decoder (Upsampling)
Skip Connections
Convolutional architecture with skip connections
Local receptive fields and hierarchical features
Proven success for early diffusion models (DDPM, Improved DDPM, etc.)
2022–2023
DIFFUSION TRANSFORMER (DIT)
Transformers for Diffusion Models
TRANSFORMER BLOCKS ...
Patch Embedding
Transformer Blocks
Patch Projection
Replaces convolutions with global self-attention
Captures long-range dependencies effectively
Scales better with model size and data
Foundation for scalable generative modeling
2024+
MMDiT (MULTI-MODAL DIT)
Unified Multi-Modal Diffusion Transformer
Text
Image
Audio
Depth
Video
Other
MULTI-MODAL TRANSFORMER BLOCKS ...
Modality Tokens
Transformer Blocks
Output Projection
Extends DiT to multiple modalities and tasks
Unified representation across diverse data types
Enables richer generation and understanding
Powers next-gen AI systems and applications
2015–2021
The Foundation
U-Net establishes the standard for diffusion models.
2022–2023
The Transformer Shift
DiT introduces transformers, unlocking scalability and global context.
2024+
The Multi-Modal Future
MMDIT unifies modalities, driving the future of generative AI.

THE 2026 FRONTIER

Two roads to the same goal

Today's leading systems split into two philosophies. Knowing the difference helps you pick the right tool — and read the headlines.

Diffusion / Flow

Iteratively denoise a latent. Champions of raw visual fidelity, open weights, and fine-grained control.

Strengths texture, photorealism, open ecosystem, editability

`Examples: FLUX.1 / FLUX.2, SDXL & SD3.5, Qwen-Image`

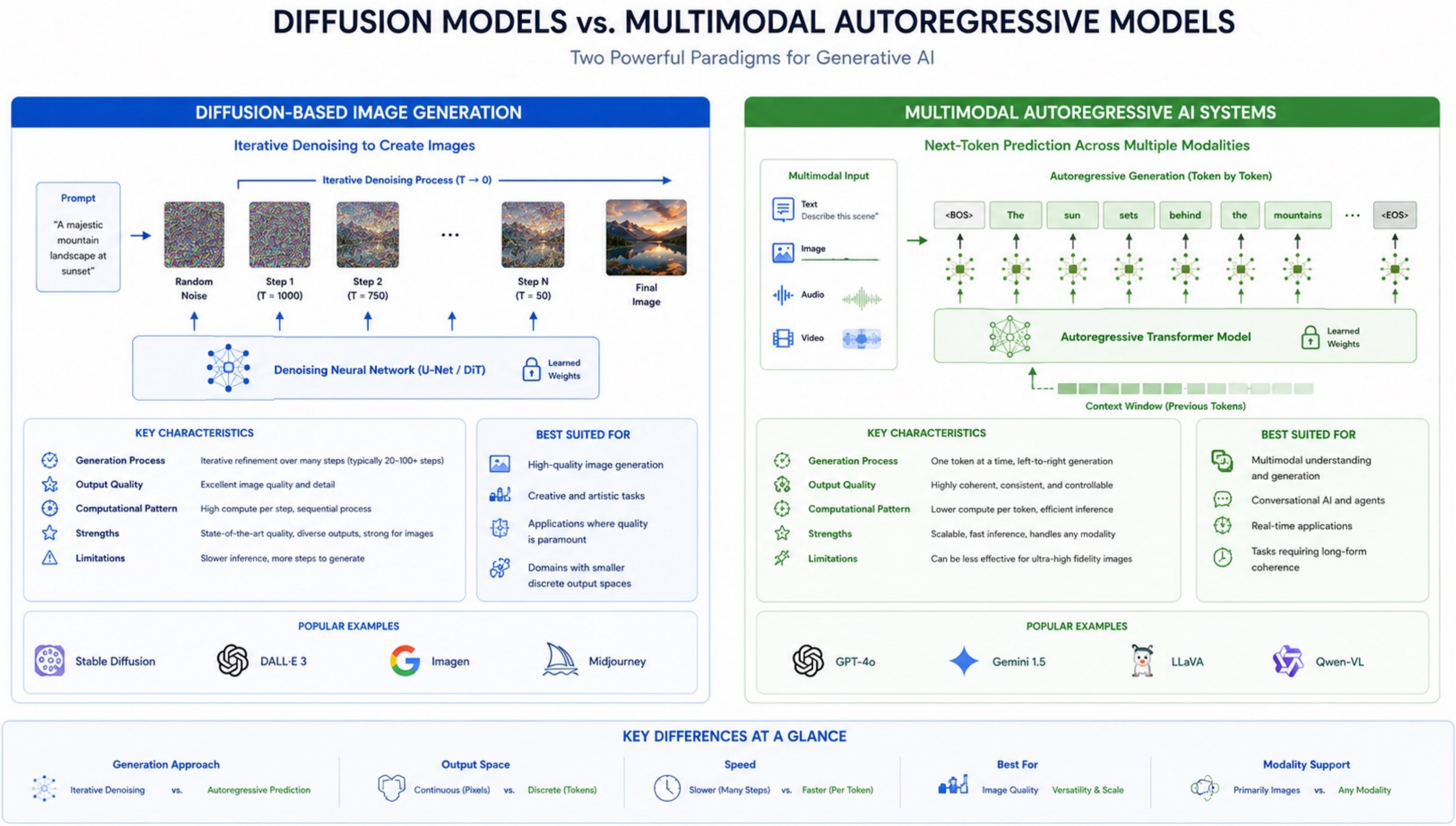
Autoregressive / Unified

Treat pixels as tokens and predict them inside one multimodal model — so it can reason about an image the way it reasons about text.

Strengths prompt-following, text rendering, world knowledge, editing

`Examples: GPT-Image series, Gemini ‘Nano Banana’ image models`

The lines are blurring — the frontier is unified multimodal systems, not lone pixel generators.


DIFFUSION MODELS vs. MULTIMODAL AUTOREGRESSIVE MODELS
Two Powerful Paradigms for Generative AI
DIFFUSION-BASED IMAGE GENERATION
Iterative Denoising to Create Images
Iterative Denoising Process (T → 0)
Prompt
"A majestic mountain landscape at sunset"
Random Noise
Step 1 (T = 1000)
Step 2 (T = 750)
Step N (T = 50)
Final Image
Denoising Neural Network (U-Net / DiT)
Learned Weights
KEY CHARACTERISTICS
Generation Process
Iterative refinement over many steps (typically 20-100+ steps)
Output Quality
Excellent image quality and detail
Computational Pattern
High compute per step, sequential process
Strengths
State-of-the-art quality, diverse outputs, strong for images
Limitations
Slower inference, more steps to generate
BEST SUITED FOR
High-quality image generation
Creative and artistic tasks
Applications where quality is paramount
Domains with smaller discrete output spaces
POPULAR EXAMPLES
Stable Diffusion
DALL·E 3
Imagen
Midjourney
MULTIMODAL AUTOREGRESSIVE AI SYSTEMS
Next-Token Prediction Across Multiple Modalities
Multimodal Input
Text
Describe this scene"
Image
Audio
Video
Autoregressive Generation (Token by Token)
<BOS>
The
sun
sets
behind
the
mountains
<EOS>
Autoregressive Transformer Model
Learned Weights
Context Window (Previous Tokens)
KEY CHARACTERISTICS
Generation Process
One token at a time, left-to-right generation
Output Quality
Highly coherent, consistent, and controllable
Computational Pattern
Lower compute per token, efficient inference
Strengths
Scalable, fast inference, handles any modality
Limitations
Can be less effective for ultra-high fidelity images
BEST SUITED FOR
Multimodal understanding and generation
Conversational AI and agents
Real-time applications
Tasks requiring long-form coherence
POPULAR EXAMPLES
GPT-4o
Gemini 1.5
LLaVA
Qwen-VL
KEY DIFFERENCES AT A GLANCE
Generation Approach
Iterative Denoising vs. Autoregressive Prediction
Output Space
Continuous (Pixels) vs. Discrete (Tokens)
Speed
Slower (Many Steps) vs. Faster (Per Token)
Best For
Image Quality Versatility & Scale
Modality Support
Primarily Images vs. Any Modality

WHERE IT PAYS OFF

From lab to industry

Marketing & e-commerce

On-brand product shots, ad variants, and localized creative at scale.

Media & entertainment

Concept art, storyboards, VFX, and rapid pre-visualization.

Design & architecture

Mood boards, façade studies, and fast iteration on visual ideas.

Healthcare & science

Image enhancement, data augmentation, and molecular design support.

Gaming

Textures, assets, and world art generated and iterated in minutes.

Education & training

Custom diagrams, illustrations, and visual explainers on demand.

AI IN ACTION ACROSS INDUSTRIES

. Transforming Workflows. Enhancing Creativity. Delivering Impact. .

MARKETING

Smarter Campaigns. Deeper Engagement.

- Audience segmentation
- Content generation
- Predictive analytics

HEALTHCARE

Better Outcomes. Smarter Care.

- Medical image analysis
- Drug discovery
- Patient risk prediction

GAMING

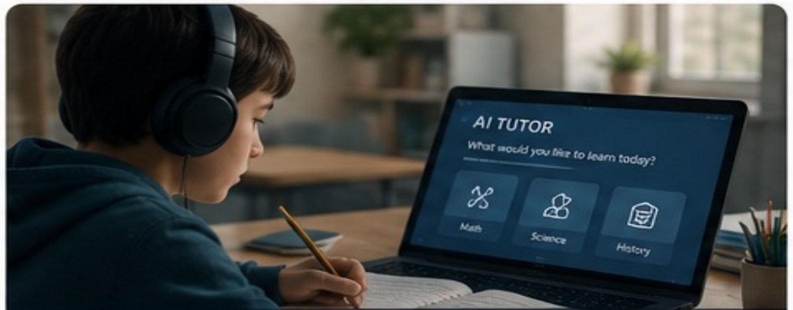
Immersive Worlds. Intelligent Gameplay.

- Procedural content generation
- NPC behavior modeling
- Real-time personalization

ARCHITECTURE

Design Smarter. Build Better.

- Generative design
- 3D modeling
- Sustainability analysis



EDUCATION

Personalized Learning. Better Results.

- Adaptive learning paths
- Automated assessments
- Intelligent tutoring systems

MEDIA & ENTERTAINMENT

Create. Enhance. Captivate.

- Content generation
- Visual effects
- Audio synthesis

AI is not just a technology—it's a catalyst for innovation across every industry and every workflow.

Increase Efficiency | Drive Innovation | Enhance Experiences | Shape the Future

THE HONEST PART

Limits, failure modes & responsible use

World-class practice means naming what these systems still get wrong — and using them responsibly.

Anatomy & fine structure

Hands, fingers, text, and symmetry still slip — improving, not solved.

Deepfakes & consent

Realistic fakes of real people demand consent, disclosure, and care.

Resolution & composition

Older models favoured fixed sizes; complex multi-object scenes can drift.

IP & provenance

Respect rights; favour watermarking and content credentials (C2PA).

Faithfulness & bias

Outputs inherit biases in training data; prompts aren't always followed exactly.

Regulation is arriving

Disclosure and transparency rules (e.g. the EU AI Act) are now in play.

PART 4

Hands-On

Two short demos to make it real — one open-source, one API. The code is in your reference notebooks.

DEMO 1 · OPEN-SOURCE

Stable Diffusion text-to-image

Run the full pipeline locally with Hugging Face diffusers — positive and negative prompts, adjustable strength.

stable_diffusion_t2i.py

```
from diffusers import StableDiffusionPipeline
import torch

pipe = StableDiffusionPipeline.from_pretrained(
    "stabilityai/stable-diffusion-2-1",
    torch_dtype=torch.float16,
).to("cuda")

image = pipe(
    prompt="a red fox in fresh snow, golden hour",
    negative_prompt="blurry, extra limbs",
    num_inference_steps=30,
    guidance_scale=7.5,
).images[0]

image.save("fox.png")
```

What to watch

1. Model loads VAE + U-Net + text encoder — Part 1 in code
2. negative_prompt steers AWAY from unwanted traits
3. steps & guidance_scale = your quality/creativity dials

DEMO 2 · HOSTED API

Image generation via the OpenAI API

When you want top quality without managing GPUs, a hosted model is one call away. Same idea — text in, pixels out.

generate_with_api.py

```
from openai import OpenAI
import base64

client = OpenAI()

resp = client.images.generate(
    model="gpt-image-1",
    prompt="a serene mountain lake at dawn, "
           "photorealistic, soft mist",
    size="1024x1024",
)

img = base64.b64decode(resp.data[0].b64_json)
open("lake.png", "wb").write(img)
```

Open- vs hosted

1. Hosted: best quality, no infra, pay per image
2. Open: full control, private, customizable, free to run
3. Same mental model — the prompt is the steering wheel

IF YOU REMEMBER NOTHING ELSE

Three ideas to carry home

Generation is denoising in reverse

Models learn to remove noise, then sculpt images out of pure static. Latent space makes it affordable.

01

Words steer pixels through a shared map

Embeddings + CLIP + contrastive learning let meaning guide every denoising step.

02

The frontier is unified & multimodal

U-Net → Transformer → one model that reasons across text, image, and beyond — used responsibly.

03

THANK YOU

Questions & conversation

You typed a sentence — and now you know how the machine paints it.

Let's discuss your use cases — and where you'd take this next.